Sashwat Tanay

Assistant Professor
Universidad de Ingeniería y Tecnología (UTEC)
Jr. Medrano Silva 165, Barranco, Lima, Peru

stanay@utec.edu.pe
sashwattanay.github.io/site
ORCID: 0000-0002-2964-7102
INSPIRE-HEP, Google Scholar

EMPLOYMENT

Assistant Professor Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru — 2026-present

Postdoctoral Fellow LUX, Paris Observatory - PSL University — 2023-2025

Adjunct Instructor University of Mississippi — 2022-2023

Teaching and Research Assistant University of Mississippi — 2016-2022

Junior Research Fellow Tata Institute of Fundamental Research, Mumbai — 2013-2015

EDUCATION

Ph.D. (Physics) University of Mississippi (Advisor: Prof. Leo C. Stein) — 2016-2022

B.Tech. (Mechanical Engineering) Indian Institute of Technology (IIT) Ropar — 2009-2013

AWARDS, FELLOWSHIPS AND FUNDING

UTEC Internal Research Grant, UTEC, Lima (100,000 PEN or $30,000 approx.) — 2026-2027

PSL Postdoctoral Fellowship, Paris Observatory – PSL University — 2023-2025

FGSA Travel Award for Excellence in Graduate Research, APS ($500) — 2022

Graduate School Honors Fellowship, Univ. of Mississippi ($12,000) — 2016-2020

Junior Research Fellowship, Tata Institute of Fundamental Research, Mumbai — 2013-2015

RESEARCH INTERESTS

• Gravitational waves • Post-Newtonian dynamics of binary black holes • Quasi-normal mode ringdown of black holes • Hamiltonian dynamical systems • Extreme mass ratio inspirals

RESEARCH SUPERVISION

Tom Colin (master's student, Ecole Normale Supérieure, Paris) led to Publication (2) — Oct 2023-Aug 2026

Manuel A. Morales (undergrad, Universidad Nacional de Trujillo) led to Publication (1) — Nov 2024-Oct 2026

Rickmoy Samanta (postdoc, ISI Kolkata) led to Publication (4) — Sep 2021-Sep 2022

Pranav Kasetty (undergrad thesis co-advisor, IISc Bengaluru) — Oct 2021-Apr 2022

TEACHING EXPERIENCE

MCI8101 Mathematics for Engineering Fall 2026
UTEC, Lima — course website

CC1105 Statistics and Probability I Spring 2026
UTEC, Lima — course website, lecture notes & student feedback

Phys 211 Physics for Science and Engineering I Summer 2023
Univ. of Mississippi — course website & lecture notes

Phys 221, 222, 223, 224 Undergrad Physics Lab Courses (Teaching Assistant) 2016–2022
Univ. of Mississippi

PROFESSIONAL SERVICE

Referee Physical Review & Physical Review Letters Feb 2023-present

INVITED TALKS & LECTURES

National University of Callao, Peru	Jan 2026
Pontifical Catholic University of Peru (Cafecitos Astronómicos)	Dec 2025
University of the Chinese Academy of Sciences	Sep 2025
Max Planck Institute for Gravitational Physics Potsdam (ACR Seminar)	Sep 2025
Max Planck Institute for Gravitational Physics Hannover (Gravitational Wave Group Meeting)	Jun 2025
University of Mississippi (Department Colloquium)	Apr 2025
IHES, Paris-Saclay University (Amplitudes and Gravitation Seminar-IHES/IPhT)	Jan 2025
York University, Toronto (Department Colloquium)	Dec 2024
Institut d'astrophysique de Paris, Sorbonne University (GReCO seminar)	Jan 2024
IISER Pune (Physics Seminar)	Jan 2024
Paris Observatory, Paris Sciences et Lettres (PSL) University (LUTH Seminar)	Sep 2023
Missouri University of Science and Technology (Department Colloquium)	Aug 2023
Northwestern University	Jul 2023
University of Illinois Urbana-Champaign (lecture workshop; lecture notes here)	Jun 2022
Montana State University (Relativity, Astrophysics and Space Science Seminar)	Apr 2022
Max Planck Institute for Gravitational Physics Potsdam (ACR Seminar)	Jun 2021
Simon Fraser University (Cosmology Seminar)	Sep 2020

OUTREACH & SERVICE

- Podcast with The Intergalactic Space Travellers Journal (Apple Podcasts, Spotify) 2026
- Contributor on Wikipedia (English & French) including various articles on physics; see here and here
- Public Talk (Q2C, From Quantum to Cosmos), UTEC, Lima 2025
- Public talk in French (Journée du LUTH), Paris Observatory 2024
- Public Talk on Astronomy - Univ. of Mississippi 2023
- Judge at The Speaker's Edge Competition - Univ. of Mississippi 2022
- Organized STEM Summer Camp - Univ. of Mississippi 2018, 2019
- Organized Spooky Physics Night - Univ. of Mississippi 2016, 2017, 2018
- YouTube videos on research and popular science

PUBLICATIONS AND RESEARCH ARTICLES

1. T. Colin, M. A. Morales, and **S. Tanay**. Orbit-averaged spinning binary black holes: Hamiltonian and Integrability, *in prep.* 2026

2. T. Colin, **S. Tanay**, and L. Bernard. Analytical Solution of Spinning, Eccentric Binary Black Hole Dynamics at the Second Post-Newtonian Order. *Phys. Rev. D*, 114:064051, 2026, arXiv:2603.20031

3. V. Witzany, V. Skoupý, L. C. Stein, and **S. Tanay**. Actions of spinning compact binaries: Spinning particle in Kerr matched to dynamics at 1.5 post-Newtonian order. *Phys. Rev. D*, 111(4):044032, 2025, arXiv:2411.09742

4. R. Samanta, **S. Tanay**, and L. C. Stein. Closed-form solutions of spinning, eccentric binary black holes at 1.5 post-Newtonian order. *Phys. Rev. D*, 108(14):124039, 2023, arXiv:2210.01605

5. **S. Tanay**. Towards a more robust algorithm for computing the Kerr quasinormal mode frequencies, 2022, arXiv:2210.03657

6. **S. Tanay**, L. C. Stein, and G. Cho. Action-angle variables of a binary black hole with arbitrary eccentricity, spins, and masses at 1.5 post-Newtonian order. *Phys. Rev. D*, 107(26):103040, 2021, arXiv:2110.15351

7. G. Cho, **S. Tanay**, A. Gopakumar, and H. M. Lee. Generalized quasi-Keplerian solution for eccentric, nonspinning compact binaries at 4PN order and the associated inspiral-merger-ringdown waveform. *Phys. Rev. D*, 105(6):064010, 2022, arXiv: 2110.09608

8. **S. Tanay**, L. C. Stein, and J. T. Gálvez Ghersi. Integrability of eccentric, spinning black hole binaries up to second post-Newtonian order. *Phys. Rev. D*, 103(6):064066, 2021, arXiv: 2012.06586

9. **S. Tanay**, A. Klein, E. Berti, and A. Nishizawa. Convergence of Fourier-domain templates for inspiraling eccentric compact binaries. *Phys. Rev. D*, 100(6):064006, 2019, arXiv:1905.08811

10. **S. Tanay**, M. Haney, and A. Gopakumar. Frequency and time domain inspiral templates for comparable mass compact binaries in eccentric orbits. *Phys. Rev. D*, 93(6):064031, 2016, arXiv:1602.03081